

Probability and Computing

Question 3

(c) Clearly, Z_t is a function of $Y_0, \dots, Y_t$.

Since $0 < f(i)^{-N(Y_t, i)} \leq 1$, $E[Z_t] < \infty$.

$$\begin{aligned} E[Z_{n+1}|Y_0, \dots, Y_n] &= Z_n \left(\sum_{i \neq j} p(Y_n, i, j) f(i)^{-1} f(j) + \Pr(\forall k \in \tau, N(Y_{n+1}, k) = N(Y_n, k)) \right) \\ &= Z_n \left(\sum_{i \neq j} \frac{f(i) N(Y_n, i) N(Y_n, j) f(i)^{-1} f(j)}{n F(Y_n)} + 1 - \sum_{i \neq j} p(Y_n, i, j) \right) \\ &= Z_n \left(\sum_{i \neq j} \frac{f(j) N(Y_n, i) N(Y_n, j)}{n F(Y_n)} + 1 - \sum_{i \neq j} p(Y_n, i, j) \right) \\ &= Z_n \left(\sum_{i \neq j} p(Y_n, i, j) + 1 - \sum_{i \neq j} p(Y_n, i, j) \right) \\ &= Z_n \end{aligned} \tag{1}$$

(d) Use a "primed" version of everything

$$f'(0) := f(0)$$

$$\forall i \in [n], f'(i) := f(1)$$

Given $Y_0, u \in [n]$, let

$$Y'_{0,u} = \begin{cases} 0 & \text{if } Y_{0,u} = 0 \\ u & \text{otherwise} \end{cases}$$

$$\tau' := \{i \in \{0, \dots, n\} \mid \exists u \in [n] \text{ s.t. } Y'_{0,u} = i\}$$

$$F'(t) := \sum_{u \in [n]} f'(Y'_{t,u})$$

π'_i is the probability that i wins. By symmetry,

$$\forall i, j \in \tau', i, j > 0, \pi'_i = \pi'_j = \pi'$$

For $i \in \{1, 2\}$,

$$\pi_i = \sum_{u \in \tau': Y_{0,u} = i} \pi'_u = N(Y_0, i) \pi'$$

$$\frac{\pi_1}{\pi_2} = \frac{N(Y_0, 1)}{N(Y_0, 2)}$$